UNIT 2 · 9 hours

Necessity and Principle of Operation of a Transformer



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Explain why transformers are necessary in power systems
- State the principle of operation (mutual induction)
- Derive the ideal voltage, current and turns ratios
- Explain what happens when a load is connected

Syllabus context

Transformers: necessity of transformers, principle of operation, types and construction of single-phase transformers, EMF equation, losses and efficiency.

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Necessity of a transformer

Power ∝ V × I, but line loss ∝ I^2R.

For the same power, raising the voltage lowers the current and the loss.

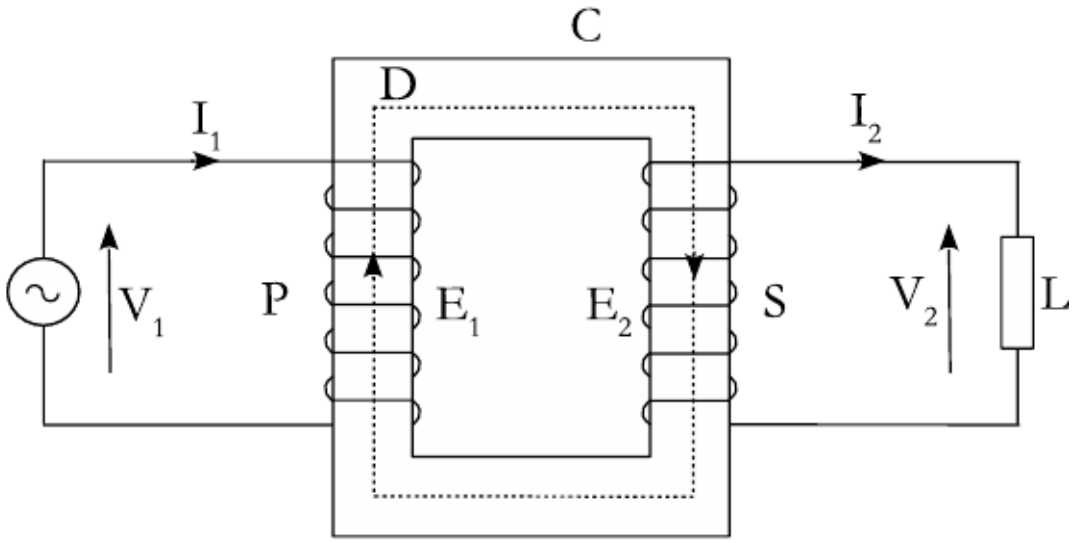
Generate at ~11 kV, step up to 132–400 kV for transmission, step down to 230/415 V for consumers.

This ability is the main reason AC replaced DC in power systems.

Other uses: isolation, impedance matching, measurement (CT, PT).

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Working principle: mutual induction



- AC voltage V1 on primary P drives alternating current.
- This produces alternating flux φ in the core.
- Flux links the secondary S: EMF E2 induced (mutual induction).
- Flux also induces E1 in the primary (self-induction).

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Ideal transformer relations

$$E_2/E_1 = N_2/N_1 = K$$

voltage (turns) ratio

$$V_1 I_1 = V_2 I_2$$

volt-ampere balance

$$I_1/I_2 = N_2/N_1 = V_2/V_1$$

current ratio

- K > 1: step-up; K < 1: step-down.
- Ampere-turns balance: N1 I1 = N2 I2 (ideal).

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Effect of loading

1

No load

Primary draws a small magnetising current to set up the flux.

2

Load connected

Secondary current I2 flows and by Lenz's law demagnetises the core.

3

Flux dips

Net flux and E1 fall slightly, so E1 < V1.

4

Primary current rises

Extra primary current restores the flux until N1 I1 ≈ N2 I2.

Flux in a transformer stays almost constant from no load to full load.

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Numerical: turns ratio

Problem A single-phase transformer has 1000 primary turns and 200 secondary turns on 230 V. The secondary load is 5 A. Find V2 and I1 (ideal).

- $K = N_2/N_1 = 200/1000 = 0.2$
- $V_2 = K\,V_1 = 0.2 \times 230 = 46$ V
- $I_1 = K\,I_2 = 0.2 \times 5$

Answer: $V_2 = 46$ V, $I_1 = 1$ A

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Key takeaways

- Transformers make efficient long-distance AC transmission possible
- Principle: mutual induction between coupled windings
- V2/V1 = N2/N1 = I1/I2 (ideal)
- Load current is balanced by extra primary ampere-turns

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Practice questions

Q1. Why is power transmitted at high voltage?

Q2. A 2200/220 V transformer has 400 primary turns. Find secondary turns.

Q3. A 5 kVA, 230/115 V transformer: find full-load currents.

Q4. Explain why a transformer does not work on DC.